HTWAVE

BEYOND LIMITS IN WIRELESS

SOFTWARE-DEFINED RADIO PLATFORM

Integrive-100

From PHY Algorithms to Real-RF Validation

Standalone 2×2 MIMO SDR with an integrated real-time PHY for wireless research and prototyping.



PRODUCT OVERVIEW

Power on. Experiment from day one.

Integrive-100 is a standalone, real-time 2×2 MIMO software-defined radio platform for advanced wireless research and rapid prototyping. With FPGA-accelerated wireless processing, ultra-low phase noise design, and a proven real-time PHY baseline, it enables high-performance wireless experimentation and rapid prototyping without external workstations, clocking infrastructure, or complex setup.

70 MHz to 6 GHz

Tuning range from VHF up to the 6 GHz Wi-Fi band

56 MHz

Instantaneous bandwidth per channel

2×2 MIMO

Two-transmit, two-receive, phase-coherent

FROM RESEARCH TO PRODUCT

Validate before you productize.

Simulation should be the beginning of validation, not the end. Integrive-100 fills the critical gap between research results and product development by enabling real-time implementation and validation on actual hardware.

Explore

Develop the idea in simulation.

PHY algorithms · MAC protocols · Routing · ISAC · Wireless applications

→

INTEGRIVE-100

Validate

Validate the idea on real hardware, in its target environment.

Real RF hardware · Multi-node networks · Industrial field tests

→

Build

Focus on product engineering.

Real-time FPGA implementation · Fixed-point · PHY/MAC integration · RF optimization · Certification

Enter product development with the core technology already validated, rather than discovering fundamental problems late.

WHAT SETS IT APART

Research-grade by Design

Truly Standalone

Core PHY runs onboard. A host PC is optional, not required for real-time operation.

Deterministic Real-Time

PHY processing runs in hardware, free from host latency and OS jitter.

Fast Bring-Up

The wireless system is ready to run without rebuilding the PHY stack from scratch.

High Signal Integrity

Low phase noise and stable RF performance for demanding wireless experiments.

Integrated PHY Baseline

A working, modular TX/RX pipeline is already in place.

Unified API

Reusable modules accessible from C, C++, and Python.

MEASURED SIGNAL INTEGRITY

≤ −35 dB EVM

TX - RX Loopback



Stable 256-QAM operation with >3 dB EVM margin.

WHAT YOU CAN BUILD

Simulation is where an idea begins. Integrive-100 is where it meets the real wireless world.

1 Unit

1× INTEGRIVE-100

PHY algorithm development

Develop and validate PHY algorithms at the symbol level over real RF.

Sync · Channel estimation · FEC · OFDM · Waveforms

N Units

N× INTEGRIVE-100 NODES

Wireless network testbed

Build a real-RF testbed for MAC, routing, coordination, and multi-node experiments.

MAC · Ad Hoc · Multi-Hop · Routing · ISAC

+ AI

INTEGRIVE-100 + AI PLATFORM

AI-enabled wireless systems

Connect to Jetson Orin or other AI/NPU platforms over USB or Ethernet.

Robotics · On-device AI · Distributed sensing · Edge intelligence

APPLICATION EXAMPLES

Take your research from algorithms to complete wireless systems.

Wireless Networking Research

Test new MAC, routing, and coordination algorithms with real wireless nodes and links.

MAC Scheduling · Channel Access · Multi-Node Coordination · Ad Hoc Networking · Multi-Hop · Routing · Topology Control

Industrial ISAC

Evaluate communication and sensing together in the intended physical environment.

1 Coordinator + N Wireless Nodes

Multipath · Metal Structures · Obstacles · Interference · Moving Objects

AI-Connected Wireless Systems

Study systems where intelligence, sensing, communication, and control operate together under real wireless constraints.

Distributed AI · Collaborative Robotics · Distributed Sensing · Edge Intelligence

Wireless Education & Research Labs

Move beyond simulation and let students build, test, and observe working wireless systems.

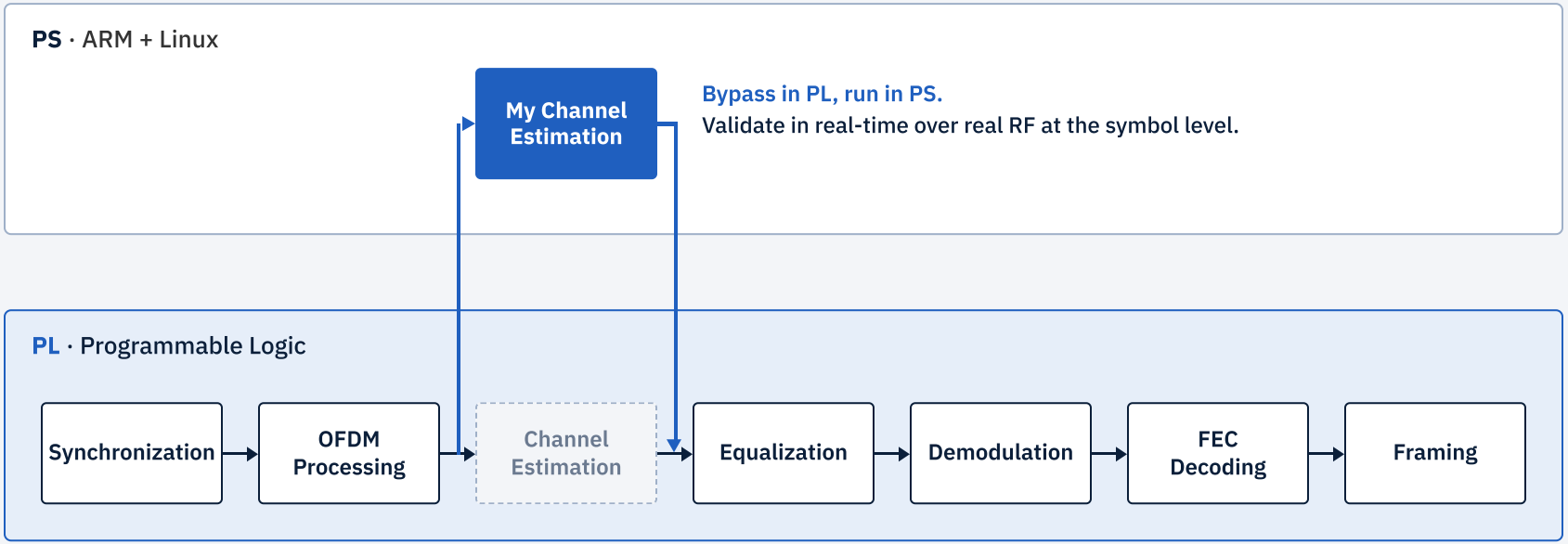
Real Packets · Interference · Mobility · Timing · Synchronization · System Integration

SINGLE-UNIT RESEARCH

Change one block. Keep the rest of the PHY running.

Integrive-100 ships with a complete real-time PHY running in PL. Bypass the block you want to evaluate and run your own implementation in the PS (ARM + Linux), while the rest of the PHY keeps running in PL.

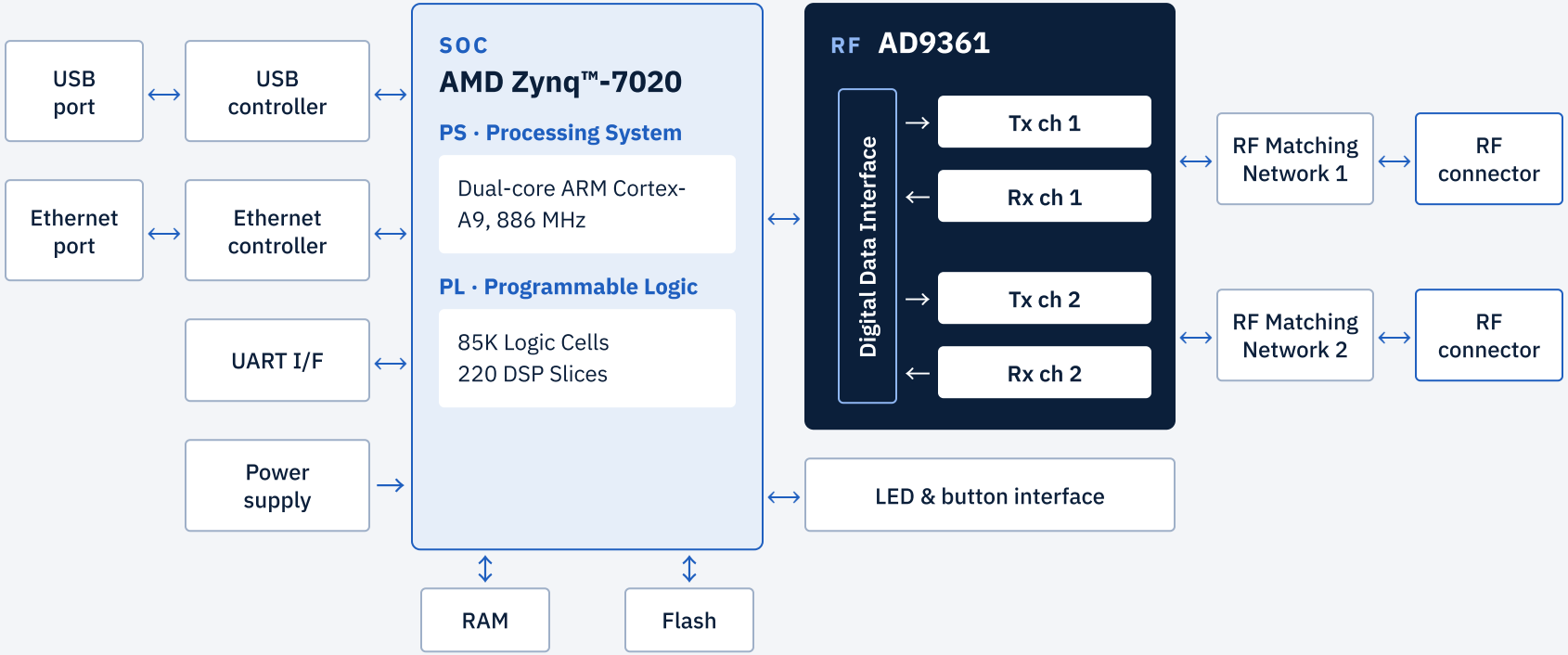
EXAMPLE: REPLACING A RECEIVE-CHAIN BLOCK



Focus on your algorithm. Integrive-100 keeps the rest of the PHY running in PL.

SYSTEM ARCHITECTURE

From antenna to algorithm, on one board.



SPECIFICATIONS

The numbers behind the platform.

RF & SDR

Architecture	**2×2 MIMO (2 TX / 2 RX)**
RF transceiver	**Analog Devices AD9361**
Frequency range	**70 MHz to 6 GHz**
Instantaneous bandwidth	**Up to 56 MHz per channel**
ADC / DAC	**12-bit, up to 61.44 MSPS**

PROCESSING

SoC	**AMD Zynq™-7020**
CPU	**Dual-core ARM Cortex-A9, 886 MHz**
Programmable logic	**85K logic cells, 220 DSP slices**
Memory	**1 Gbit DDR3 SDRAM, Flash**
Interfaces	**USB · Ethernet · UART**
Operation	**Standalone, integrated FPGA DSP**

PHYSICAL

Dimensions	**85 × 80 × 18 mm**
Weight	**Approx. 80 g**



SOFTWARE & PACKAGE

LAUNCH PRICE **Save $352**

Unbox, Power On, Start Experimenting.

MSRP ~~$1,450~~ **$1,098**

Crowdfunding special until Oct 22, 5:00 pm PDT

SOFTWARE STACK

Firmware, reusable PHY modules, APIs, and development examples are provided in an integrated environment for wireless research and rapid prototyping.

On GitHub

Manual ↗

Examples ↗

IN THE BOX

- 1× **Integrive-100 board**
- 1× **Protective case**
- 2× **RF SMA antennas**

INCLUDED ANTENNAS

Frequency	**902 to 928 MHz**
Connector	**SMA Male**
Impedance	**50Ω**
Max gain	**2 dBi**
Polarization	**Vertical**
Pattern	**Omnidirectional**
Max input power	**5 W**
Length	**172 mm**

Operate from 70 MHz to 6 GHz by changing the antenna to match the target frequency (SMA).

ABOUT US

HTWAVE

HTWAVE is a deep-tech wireless company developing advanced communication systems by combining proprietary ultra-reliable, low-latency communication technologies with SDR hardware, software, and real-time PHY expertise.

ENGINEERING SUPPORT

PHY implementation · FPGA development

Protocol integration · RF/system optimization

Customized wireless-system development

HTWAVE

BEYOND LIMITS IN WIRELESS

CONTACT

Request a quote

LAUNCH PRICE · INTEGRIVE-100 KIT

Crowdfunding special until Oct 22, 5:00 pm PDT

$1,098

Back it on Crowd Supply ↗

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